3Laws

Release beta

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Chapter 1

Introduction

The 3Laws Robotics supervisor is a software package that provides a control filter to insure safety without compromising performance. In addition, the supervisor contains a diagnostic module to monitor relevant metrics for safety and performance.

In order to make the configuration and the usage of the supervisor easier, we provide a control panel, that will guide you through the configuration of the supervisor and help you visualize the supervisor's safety filter in action.

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Chapter 2

Installation

The Supervisor is distributed as a debian package. A package is available for each of the following configuration

Ubuntu Distribution	ROS1 version	ROS2 version	Architecture
22.04	N/A	Humble/Iron	amd64/arm64v8
20.04	Noetic	Galactic/Foxy	amd64/arm64v8
18.04	Melodic	N/A	amd64/arm64v8

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2.1 Quick Installation

Copy and paste the following command in a terminal to install the Supervisor:

```
bash <(curl
https://raw.githubusercontent.com/3LawsRobotics/3laws/master/rdm/install.sh)
```

This will install the Supervisor for the current distribution and architecture and the control panel.

At the end of the installation, the control panel will be started and you will be able to start the configuration of the supervisor. To do so open a web browser and go to the following address: http://localhost:8000

2.2 Non Interactive Installation

If you want to install a specific package in a non interactive mode, here is the process:

- Download the script:

```
wget https://raw.githubusercontent.com/3LawsRobotics/3laws/master/rdm/install.sh
```

- Make it executable

```
chmod +x install.sh
```

- Run it with your arguments

```
sudo ./install.sh [-hyf] [-r <ROS_DISTRO>] [-a <ARCH>] [-v <UBUNTU_VERSION>]
```

if `-yf -r <ROS_DISTRO> -a <ARCH> -v <UBUNTU_VERSION>` specified, the script is non interactive

Chapter 3

Quick Start

Before starting the supervisor be sure to have your ROS environment correctly set up and sourced.

```
source /opt/ros/<DISTRO>/setup.sh
```

To launch the Supervisor you can use the following command:

```
ros2 launch lll_rdm rdm.launch.py
```

You can add the following command line options:

- **use_sim_time**: <Boolean, default: false> Specify if the RDM use the ros time or the machine time
- **config_filename**: <String> Path to the config file if not in the */opt/3lawsRoboticsInc/robot_diagnostic_module/config* folder.
- **robot_id**: <String> name of your robot
- **log_level**: <trace | debug | info | error, default: info> Log level of the RDM
- **dry_run**: <Boolean> Start the RDM, check the config file and turn off without error if everything has been launched correctly

By default the supervisor will use the ros time and the config file in /opt/3lawsRoboticsInc/robot_diagnostic_module/config/rdm_config.yaml. You can change this default folder by setting the environment variable *LLL_CONFIG_FOLDER* to the desired path.

Chapter 4

Control Panel

The **Control Panel of 3LawsRobotics** is a helper tool to help the configuration of the 3Laws Robotics supervisor. It provides a web interface to configure the 3Laws Robotics supervisor and visualize the robot's state. Currently in alpha version, it runs on your local network and can be accessed from any device with a web browser. The backend is written in Python and uses the CGI protocol (Common Gateway Interface) to communicate with the web interface.

In order to get your account setup and use freely this RDM, please contact support@3lawsrobotics.com

This product is still at a alpha development stage. New features are developed every month and some breaking changes can still happened. The 3LawsRobotics Team is available to answer your question.

For now only a ROS1, ROS2 interfaces are configurable via this control panel.

4.1 Launch the Control Panel

The control panel is installed as a systemd service. To start the control panel, use the following command:

```
sudo systemctl start lll_control_panel
```

and to stop it:

```
sudo systemctl strop lll_control_panel
```

Also you can start it manually with the following command:

```
python3 /opt/3lawsRoboticsInc/control_panel/server.py
```

You can add the following command line options:

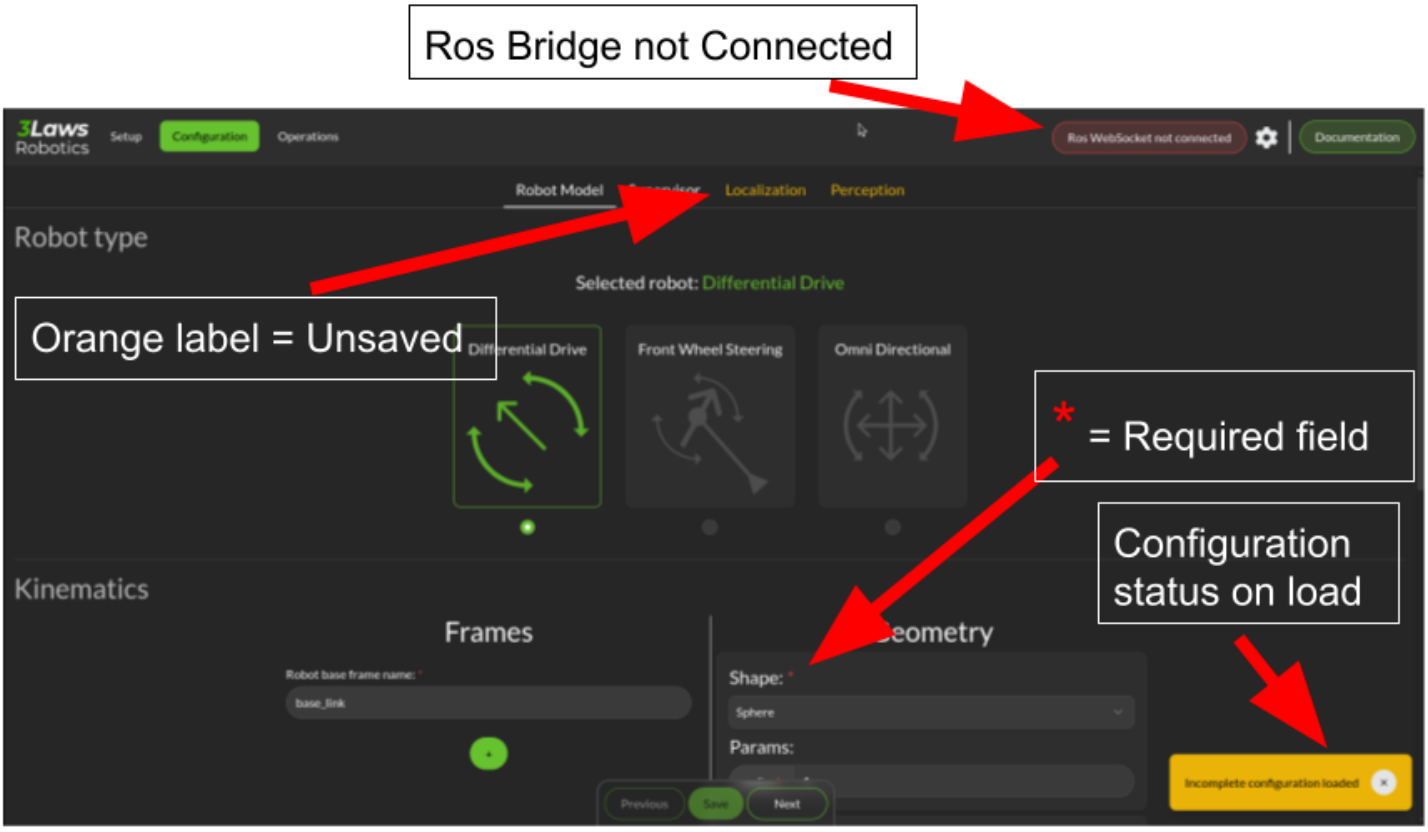
- **-p or –port**: <number, default: 8000> Specify the port to use

You can specify the emplacement of the configuration file by setting the environment variable *LLL_CONFIG_FOLDER*

4.2 Use the Control Panel

The control panel is meant to be user friendly and safe explanatory. However some detail might be useful to know.

Here is a picture of the control panel:

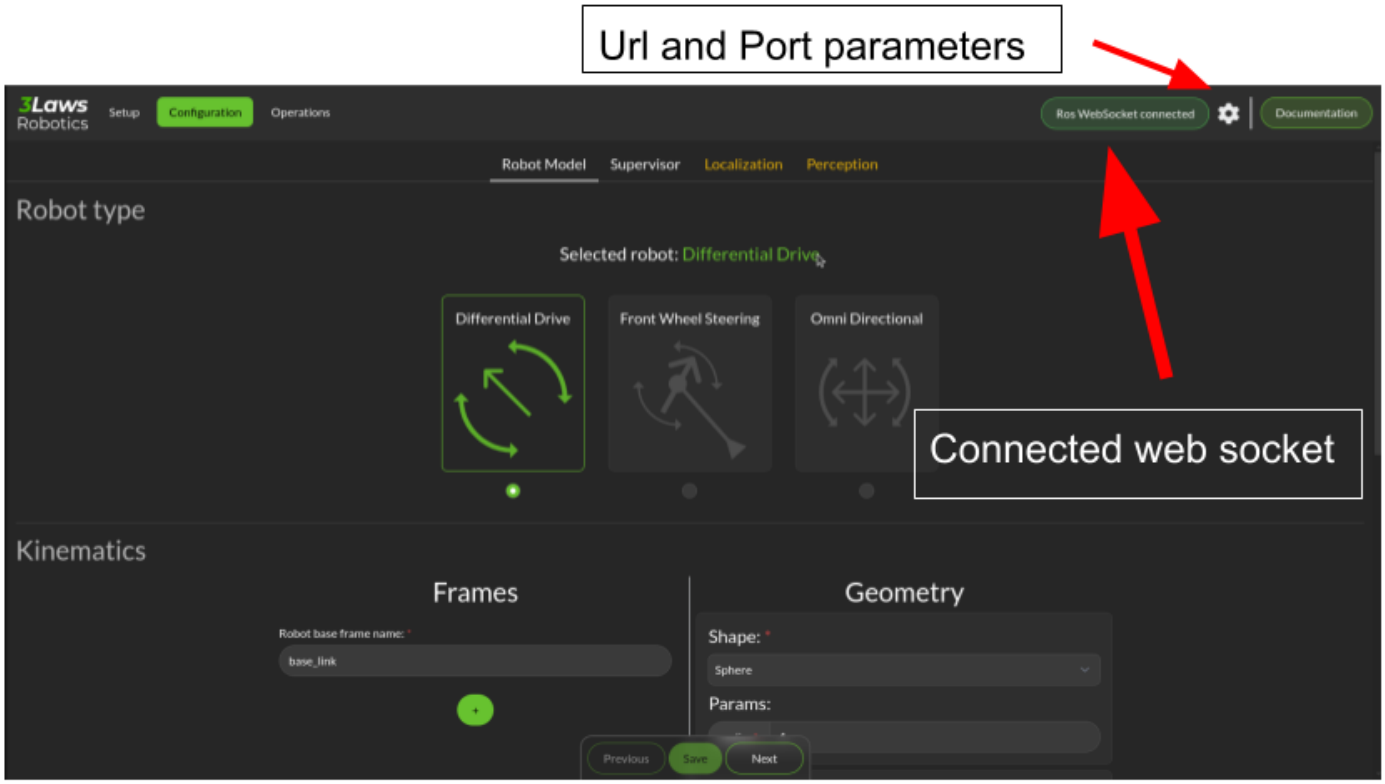


The control panel is divided in 3 parts:

- The Navigation bar, where you can find the main pages, a Ros Bridge WebSocket status indicator and a WebSocket parameter button.
- The main page, where you can find the configuration form and the control buttons.
- The footer, where you can find the version number and the 3LawsRobotics website link and a link to the update pages.

In order to use this control panel at its best, you need to configure the Ros Bridge WebSocket parameters. To do so, click on the WebSocket parameter button in the navigation bar and fill the form with the correct parameters.

By default, the control panel is configured to connect to a Ros Bridge WebSocket running on the same machine, on the port 9090. More information on the Ros Bridge WebSocket can be found following this link: Ros Bridge WebSocket.



A successful connection to the Ros Bridge WebSocket will be indicated by a green status indicator in the navigation bar.